

Chapter 2 Statistical Learning

2.1.1 Why estimate F?

1. To make prediction

2. To understand the relationship:

Sometimes, we care less about prediction and more about insight

-How does the salary depend on the experience?

-Which feature matter most?

Estimate the F help explain how X affect Y

3. Because the noise exist

Real world data is messy even with the same X, output vary. So, instead of learning noise, we aim for a function that capture the pattern, not randomness.

How can we estimate the F?

Step 1:

Choose a form of F (2 Approaches)

-Parametric method

Assume a specific shape for F

Ex: Linear regression

$$Y = B_0 + B_1X$$

Pros:

-Simple - fast - easy to interpret

Cons:

-can be too rigid (simple) (Underfitting)

-Non parametric method

do not assume a fixed form

Ex: kNN, neural network, Decision tree, Random forest

Pros:

-flexible - can model complex pattern

Cons:

-need more data - can overfitting

Step 2:

Defines loss function:

This measures how wrong our estimate is

Ex: -Mean Square Error (regression) - Miss-entropy loss (Classification)

Step 3: (Learn from data)

We find $\hat{F}$ that minimize the loss

This can be done by gradient Descent and Closed-form Solution

2.2 Assessing Model Accuracy

2.2.1 Assessing the model accuracy

A model that work only on training data is useless

we care about: how well the model performs on unseen data.

--> Always evaluate models using *test error*, not *training error*

2.2.2 Bias-Variance trade-off

Why test error exist?

Even the best model make mistake because:

- Wrong assumption (Bias)
- Sensitivity tp data (Variances)
- noise (irreducible error)

Bias = Error caused by oversimplifying the real relationship.

characteristic:

- Model too rigid
- Misses imported pattern
- lead to underfitting

Ex: using a straight line to fit a curve pattern.

Variance = error caused by model sensitivity to training data.

Characteristic:

- Model change a lot with small data change
- fit noise
- lead to overfitting

Irreducible error:

- Error caused by noise in data
- cannot be eliminated
- set a lower bound on test error

Mathematic composition

Expected test MSE = Bias **2** + **Variance** + **Irreducible error**

2.2.3 Classification setting

Classification error rate

Error rate = **incorrect prediction**/**Total prediction**

Model sensitivity in classification

+Simple classifier

-High bias

-Low variance

+Complex classifier**

-low bias

-High variance

Ex:

-K-NN with large K --> Underfitting

-k-NN with small K --> Overfitting